Devesh P.K.

LinkedIn | deveshpk GitHub | dev1237 Email | deveshpkk@gmail.com Phone | +91 82209 47118

Network security and infrastructure engineer with packet-level experience across TCP/IP, TLS, DNS and BGP — from DPI and middlebox analysis at IIT Bombay to high-availability service deployment on RHEL. CCNA in progress.

Education

SVNIT Surat – B.Tech, Electronics & Communication Engineering Surat, India | Nov. 2024 – Apr. 2028
Relevant coursework: Computer Networks, Cloud Computing *(both ongoing)*
CBSE – Class XII 85.6% (2023) • Class X 84.6% (2021) Bahrain

Experiences & Internships

•IT Infrastructure Intern (Incoming) *Dec. 2026 – Jan. 2027*
Gulf Business Machines (GBM) *Seef, Bahrain*

– Incoming winter placement across **enterprise networking, Linux server platforms and cloud infrastructure** delivery

•Internet Censorship Measurement & Network Adversary Localization (UAE) *May 2026 – July 2026*
Trust Lab, INDIAN INSTITUTE OF TECHNOLOGY (IIT) BOMBAY *certificate* | *report*

– Engineered a **3-VM middlebox testbed** (client–censor–server) emulating on-path censorship via Linux **NFQUEUE** across **HTTP Host, TLS SNI, DNS and STUN** (DROP / RST / block-page / forged-answer)
– Ran a **nation-wide UAE measurement**: 6.03M IPs → yarrp (**593,043** modified-path) → ZMap → **509 confirmed censored** hosts across **four protocols**; characterized the DPI as **stateless in the measured set**
– **Attributed the censoring middlebox**: TTL-based injector localization despite **source-spoofed resets**, off-path vs. in-path classification, and an **IP-ID side-channel** identifying **two co-located injectors** at the Etisalat (AS8966) backbone
– Packaged the whole pipeline as **phantomprobe** | *link* – a **portable single-binary open-source C tool** (libpcap only) that detects censorship, bounds **residual-filtering duration**, and attributes the middlebox from any vantage

Achievements & Distinctions

•Critical Vulnerability Discovery on Institutional Server Infrastructure (Documentation) *July 2025*
SVNIT Surat - Faculty-Approved Penetration Testing Initiative

– Discovered and responsibly **disclosed multiple high-severity vulnerabilities in institutional web infrastructure**, earning faculty recognition and official authorization for a comprehensive penetration test
– Ran vulnerability enumeration over **LDAP, SMB, FTP, and DNS** (Metasploit), **extracting password hashes and sensitive organizational metadata**
– Delivered detailed findings and remediation recommendations that **improved the institutional security posture**

•Highly-Available SSH Gateway (RHEL) — Nginx L4 Load Balancing & Keepalived/VRRP | **link** *Aug. 2025*
Two-node active/active deployment, University Electronics Dept

– Configured **Nginx stream (Layer-4) load balancing** proxying **TCP/22** across two backends with **round-robin** distribution and passive health checks (`max_fails=3`, `fail_timeout=30s`) – verified as an exact **4:4 split** across 8 concurrent sessions – **eliminating 100% of server crashes** for **150**+ concurrent SSH users and **doubling capacity to 300**+ sessions
– Eliminated the load balancer as a **single point of failure** by running **keepalived/VRRP** on **both** nodes behind a **floating virtual IP**
– If either node goes down the other takes over **automatically in under 4 seconds**, with **no interruption to active SSH sessions** – verified by both a graceful shutdown and a forced kill of the live node

Technical Skills & Tools

- **Security:** Network Security & VAPT, Network Forensics & Traffic Analysis, DPI / Middlebox Analysis, OSINT, Log Analysis, Incident Response
- **Measurement / packet tooling:** yarrp, ZMap, raw sockets, libnetfilter_queue, Scapy
- **Platforms:** Linux (RHEL/Ubuntu/Kali), Windows, macOS, Nginx
- **Protocols:** TCP/IP, TLS, DNS, BGP
- **Standards:** ISO 27001, HIPAA, GDPR
- **Focus & Interests:** Network Security & Secure Architecture, Network VAPT, High Availability & Resilient Infrastructure, Network Measurement & Topology Inference, SOC & Incident Response, Network Engineering
- **Offensive tooling:** Metasploit, Nmap, Burp Suite, Wireshark, Gobuster, John the Ripper, OWASP ZAP

Certifications

- **CCNA** – Cisco Certified Network Associate *(ongoing)*
- Cyber Job Simulations – **Deloitte** (log analysis) & **Commonwealth Bank** (incident response, security awareness, penetration testing)
- **TCM Security** Linux 100: Fundamentals • **EC-Council** Deep Web & Cybersecurity

Leadership & Positions of Responsibility

•Core Member, Research & Innovation Affairs Council (RIAC), SVNIT *Feb. 2025 - Present*
– POC for a startup founder; established connections with Oracle and Samsung R&D leaders